SPIDER / on the M-VAVE SMK-37

Quick reference guide · branch spider2 (beta), 11 November 2026 · the SMK's names first, the FM-1's after them

A new project's tracks (web theme)

T1	T2	T3	T4	T5	T6	T7	T8
kit	mono	poly	acid	fm	sample	poly	mono

8 KNOBS

K1	**Volume** (the FM-1's MASTER); the screen shows the level
K2	SELECT · the button's next page; in a list, up / down
K3	ALGORITHM · picks the track: T1–T8, COND, THROW, WEAVE
K4	PRESETS · the next preset, drum key, effect, pattern or scene
K5–K8	KNOB 1–4 · the four values along the bottom of the screen
On Steps (SEQ)	
Nothing held	K5 pattern · K6 length · K7 scale · K8 groove
A pad held	K5 velocity · K6 length · K7 chance · K8 pitch
Pad + ARP	K5 trick 1 · K6 its value · K7 trick 2 · K8 its value

4 FADERS + 2 WHEELS

Faders 1–4	Levels of T1–T4; after **FADER BANK** (lit), T5–T8. A fader acts once it moves
Pitch wheel	Bends the track the keys play, up to a fifth (±7 semitones)
Mod wheel	GLOBE, Setup, MOD WHEEL: **filter**, that track's cutoff (DYNAMIC: hi cut), picked up at its value; or **CONTROL ALL**: cutoff, LFO, send I, send II or decay on every synth track at once. KNOB BANK within 5 s puts them back

37 KEYS

F3–G5	The FM-1's 27 keys, with velocity (GLOBE, Setup: KEY VELOCITY). **White 1–16** are their white keys from F3, **black 1–11** their black keys
Outer 10	C3–E3 and G♯5–C6: more notes on the same track; on a kit, its keys
On a kit	Black 1–11: its eleven drum sounds (black 1 the kick)
Arrange	White: scenes 1–16 (OCT+: 17–32). Black: mutes
Throws	FX, K2 to Throws. Hold a white key, F3–E4 on the drums, F4–E5 on the melodic tracks: mute · stutter · repeat · echo · build · floor · jumble. Black keys: warble · pan · octave · chop · ring

BUTTONS

SMK-37	FM-1 name · what it does
PATCH	HOME · Arrange; again: Song. Held: a shift (undo, redo, tempo). To a question: no
BT	SAVE · saves now (lit: unsaved). Held + white 1–2: to that slot. On a sound page, held + a white key: keep the sound
PARA	EDIT · the sound: Engine, Play, Play 2 (K2 turns the pages)
CHORD	ENV · Envelope, Filter, Filter envelope
FX	FX · Character, Sends, FX I, FX II, Throws
SCALE	SEL · the page's settings and actions. To a question: yes
GLOBE	GLO · Mix, Master EQ, Master color, Master out, LFO, Setup
ARP	ARP · Players (arp, strum, hold, random), Weave. A step's pad held: its two tricks
SEQ	SEQ · Steps; K2: Bar. Held on Steps: the keys play, and set what a pad writes
PLAY	PLAY · start / stop. SEQ + PLAY on Steps: from this bar
REC	REC · REC + PLAY: record live. Held, playing: records while held. Tapped, stopped, on Steps: step record. **Twice:** a counted-in take
NOTE REPEAT	LFO · tapped: note repeat on / off (lit). Held + pads 1–8: its rate. Held and let go alone: the LFO page
KNOB BANK	The pads: on Steps, cells ↔ live pads (lit); elsewhere, opens Steps
FADER BANK	The faders: T1–T4 ↔ T5–T8 (lit)
< / >	The bar before / after, on any page, when let go
< + >	**DAW mode** on / off: the second pressed while the first is held
OCT– / OCT+	The keys' octave. A step's pad held: nudge it. SEQ held on Steps: the pattern an octave. Arrange: scenes 1–16 / 17–32. Step record: back / rest

QUICK MOVES

Undo / redo	PATCH + REC / PATCH + PLAY
Tempo	PATCH held + turn K2, on any page
Write a note	Hold SEQ, play it, let go, tap a pad
Lock a knob	Hold a pad, press PARA, CHORD or FX, turn
New scene	PATCH, then an empty white key
Solo	Hold the track's top pad; GLOBE (Mix): hold its black key
Help	Hold any button 1 s. Recalibrate: OCT– + OCT+ at power-on

16 PADS

On Steps and Bar (SEQ): the bar's 16 cells

1	2	3	4	5	6	7	8
9	10	11	12	13	14	15	16

Track colour: notes dim, playing bright, held white. Hold + a later pad: length

Every other page (or KNOB BANK on Steps): live pads

T1	T2	T3	T4	T5	T6	T7	T8
D1	D2	D3	D4	D5	D6	D7	D8

Top: T1–T8: tap mutes (dark), hold solos, two taps keep the solo. **Bottom:** the kit's first 8 sounds, with velocity, whatever track is selected

NOTE REPEAT held: the top row sets the rate

1/4	1/8	1/8T	1/16	1/16T	1/32	1/32T	1/64

In use: bright. A pick turns repeat on: a held drum pad repeats

DAW MODE (< + >)

Keys	CHANNEL (1), C3 = 48, OCT ±3; wheels bend, CC 1
K1–K8	CC 70–77 / 102–109; faders 20–23 / 24–27 (the BANKs)
Pads	Ch 10: bottom 36–43, top 44–51; both BANKs: 52–59, 60–67
Buttons	CC, 127 / 0: < 110 · > 111 · SEQ 112 · FX 113 · PARA 114 · PLAY 115 · REC 117 · ARP 118 · NOTE REPEAT 119 · SCALE 85 · CHORD 86 · GLOBE 87 · BT 88
PATCH	Tapped: LOCAL. Held: K5 CHANNEL · K6 map · K7 ABS / REL · K8 velocity · OCT+ map editor
REAPER	K6's third map: a Mackie Control surface on USB port 2. **Over the page**

SPIDER'S SOUNDS

Engines	poly (Juno-106) · mono (SH-101) · fm (6-op) · acid (303) · sample · slice · kits (acoustic, 909, analog …) · MIDI. Factory samples built in
Rhythms	A kit: SEQ, SCALE, RHYTHM; K3 picks one of 24, SCALE writes it (+ KIT: its kit too)

SPIDER / REAPER mode

Reverse · DAW mode, CC MAP on REAPER: a Mackie Control surface for REAPER · branch spider2 (beta), 6 November 2026

Try it in the emulator first: < + >, PATCH held + K6 to REAPER, then **Pretend REAPER**

TURN IT ON

1	**< + >**: DAW mode (SPIDER's sequencer stops)
2	**PATCH held, K6** (CC MAP): DEFAULT → USER → **REAPER**. Kept, DAW mode too
3	**Waiting for REAPER** until it answers; every 3 s SPIDER asks again
Port 1	**SPIDER SMK-37**: keys, drum pads (ch 10), wheels, NAV CCs (ch 16), clock
Port 2	Mackie Control both ways: **MIDIIN2 / MIDIOUT2 (SPIDER SMK-37)** on Windows, **SPIDER DAW** on a Mac

8 KNOBS

MIX	Sends	In REAPER (KNOB BANK unlit)
K1–K8	V-pot CC 16–23	Pan of strips 1–8; flipped (FX): volume
NAV	**Sends**	**In REAPER (KNOB BANK lit)**
K1	Jog CC 60	The edit cursor; PARA lit: scrubs
K2	CC 102 · ch 16	Learn *View: Scroll horizontally*
K3	CC 103 · ch 16	Learn *View: Zoom horizontally*
K4	CC 104 · ch 16	Learn *View: Scroll vertically*
K5	CC 105 · ch 16	Learn *View: Zoom vertically*. K2–K5: mode **Relative 2**
K6	CC 106 / 107	Learn *Track: Go to previous / next track* (one CC a detent)
K7	REW 91 / FF 92	The marker before / after: nothing to learn
K8	UNDO 81 / CC 111	Left: undo. Right: learn *Edit: Redo*

4 FADERS + 2 WHEELS

Faders	Volume of strips 1–4 (pitch bend, ch 1–4); FADER BANK lit: 5–8. Flipped: pan
Pick-up	A fader waits until it crosses REAPER's level (the screen: *up* or *down to pick up*): nothing jumps. Again after a new bank, or when REAPER moves the track
Wheels	Port 1: pitch bend and CC 1, as DEFAULT

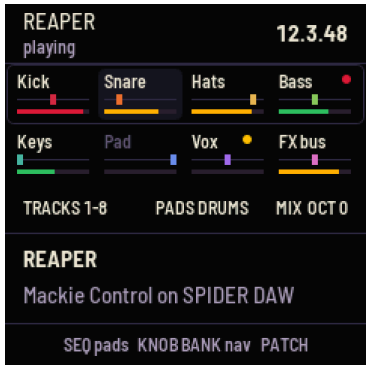
BUTTONS

SMK-37	Mackie note	In REAPER · lit when
PLAY	PLAY 94 · STOP 93	Play; while playing, stop · playing
REC	RECORD 95	Record · recording
NOTE REPEAT	CYCLE 86	Loop on / off · loop on
ARP	CLICK 89	Metronome · click on
FX	FLIP 50	Flip: faders on pan, knobs on volume · flipped
PARA	SCRUB 101	K1's jog scrubs · scrub on
SCALE	MARKER 84	Add a marker
CHORD	UNDO 81	Undo · a flash
BT	SAVE 80	Save the project · 1 s
< / >	BANK 46 / 47	Strips –8 / +8, when let go
GLOBE + < / >	CHANNEL 48 / 49	Strips –1 / +1
SPIDER's own (nothing sent)		
SEQ	The pads' page: DRUMS → MIXER → ACTIONS · lit off DRUMS	
GLOBE	Held: shift. MIXER's pads arm and solo; < > by 1	
KNOB BANK	The knobs: MIX ↔ NAV · lit on NAV	
FADER BANK	The faders: strips 1–4 ↔ 5–8 · lit on 5–8	
PATCH	Tapped: LOCAL, the keys and drum pads play SPIDER too (lit). Held: K5 CHANNEL · K6 CC MAP · K7 ABS / REL · K8 key velocity	
OCT– / OCT+	The keys' octave, ±3; both: 0	
< + >	DAW mode off: SPIDER plays again	

WHEN IT DOESN'T WORK

Still waiting	The surface needs port 2 in and out, and port 2 left off in MIDI Devices. CC MAP on REAPER?
Knob does nothing	NAV's K2–K6 and K8 right only work once learnt
Fader does nothing	It's waiting to pick up: move it past REAPER's level
CCs in a take	The track records all channels: set ch 1 or 10
Wrong tracks	< > bank by 8, GLOBE + by 1; the screen's TRACKS says which

THE SCREEN



1. Transport (**recording** red), loop, click; the time
2. Strips 1–8: name, pan, meter; the faders' four framed
3. **Filled** selected · **faint** muted · red dot armed · amber soloed
4. TRACKS · PADS page · MIX or NAV · OCT; the last control; hints

16 PADS: SEQ TURNS THE PAGE

Page	Top row, 1–8	Bottom row, 9–16
DRUMS	Notes 44–51, ch 10, port 1	Notes 36–43; LOCAL: SPIDER's kit too
MIXER	Select (white) **24–31**. GLOBE: arm (red) **0–7**	Mute (red) **16–23**. GLOBE: solo (amber) **8–15**
ACTIONS	F1–F8 **54–61**	View buttons **62–69**

MIXER: bright on (blinks as REAPER does), dim off, dark no track. ACTIONS: learn each to any action

SET UP REAPER (ONCE)

1. **Preferences → Audio → MIDI Devices:** SPIDER SMK-37 in on (and *for control messages*), out on. Port 2: leave off here
2. **Control/OSC/web → Add → Mackie Control Universal:** in and out port 2. *Map F1–F8 to markers* off. The screen leaves *Waiting*
3. **Actions → Show action list:** pick an action, Add..., turn the NAV knob or hit the ACTIONS pad
4. **Recording tracks:** input SPIDER SMK-37, ch 1 (keys) or 10 (pads), so NAV's ch 16 stays out

SPIDER IN TIME WITH REAPER

Out of DAW mode, CLOCK on receive. In REAPER, the SPIDER SMK-37 output: *Send clock* (and SPP). Play from anywhere: SPIDER's tracks start at the step they would be on

Mackie numbers are decimal, on port 2, ch 1. SPIDER's side: app/smk_mcu.h · REAPER's: reaper_csurf/csurf_mcu.cpp · more: SMK37.md, The REAPER profile

Two-sided: flip on the short edge